

```
import numpy as np
def runge_kutta_second_order(f, x0, t0, dt, num_step):

    t = [t0]
    x = [np.array(x0)]

    t_1 = t0
    x_1 = np.array(x0)

    for _ in range(num_step):
        k_1 = dt*f(t_1, x_1)
        k_2 = dt*f(t_1 + dt, x_1 + k1)
        t_2 = t_1 + dt

        x_2 = x_1 + 0.5 * (k_1 + k_2)
        t_2 = t_1 + dt

        t.append(t_2)
        x.append(x_2)

        t_1 = t_2
        x_1 = x_2

    return t,x
```